

Estimating Earnings Based on Demographic and Job- Related Information

DSC 680 – Project 2, Milestone 3

Xhoi Shyti

Business Problem

Identifying the key factors behind income disparities continues to be a major priority for educators, employers, and policymakers. It is essential for these stakeholders to gain evidence-based insights into the demographic and job-related variables that most accurately indicate whether a person earns more or less than \$50,000 annually. This study leverages machine learning techniques to classify income levels, with the goal of informing public policy, promoting equity, and guiding workforce development strategies.

Background

While income prediction has traditionally been explored within the realm of social sciences through inference-based methods, advancements in machine learning have enabled a transition toward predictive modeling. The Adult Income dataset, based on the 1994 Current Population Survey conducted by the U.S. Census Bureau, provides demographic and employment information that serves as a basis for examining how variables such as age, education level, and weekly work hours relate to income levels.

Data Explanation

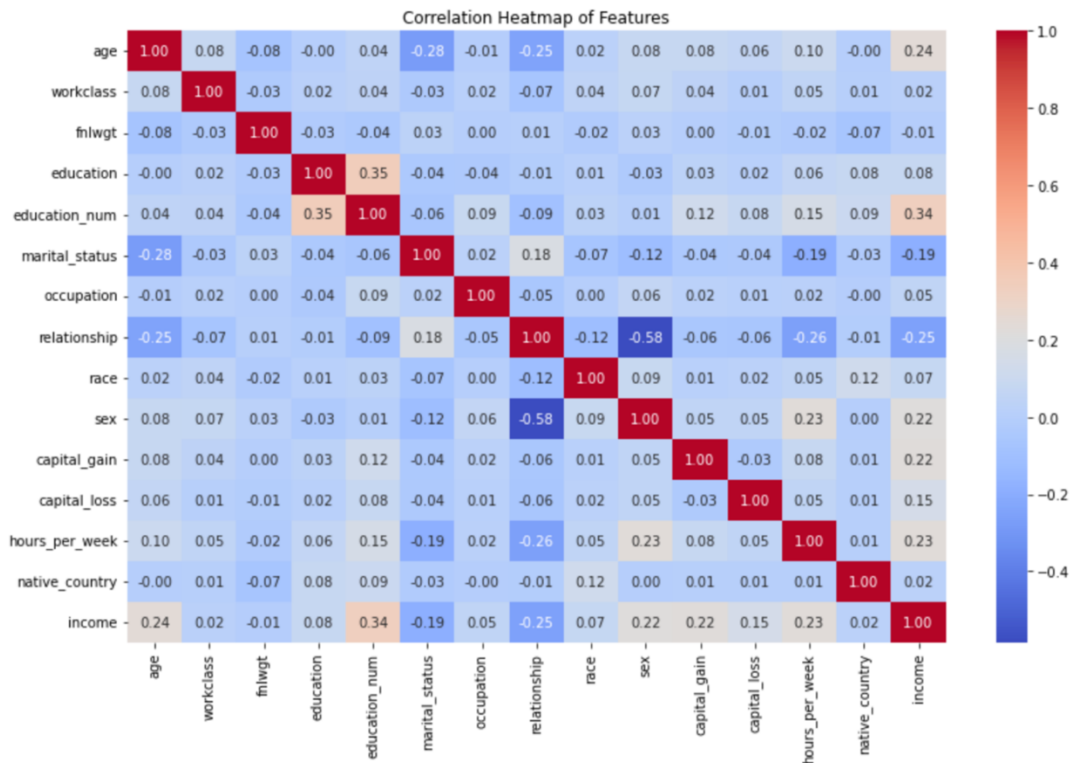
The dataset contains 32,561 records and 15 attributes, including age, type of employment, education level, numerical education score, marital status, job type, family relationship, race, gender, capital gains, capital losses, weekly working hours, country of origin, and income category. The data preparation process included the following steps:

- Addressing incomplete data and outliers
- Transforming categorical features through label encoding
- Scaling numerical variables to a standard range
- Splitting data into training and test sets

Exploratory Data Analysis

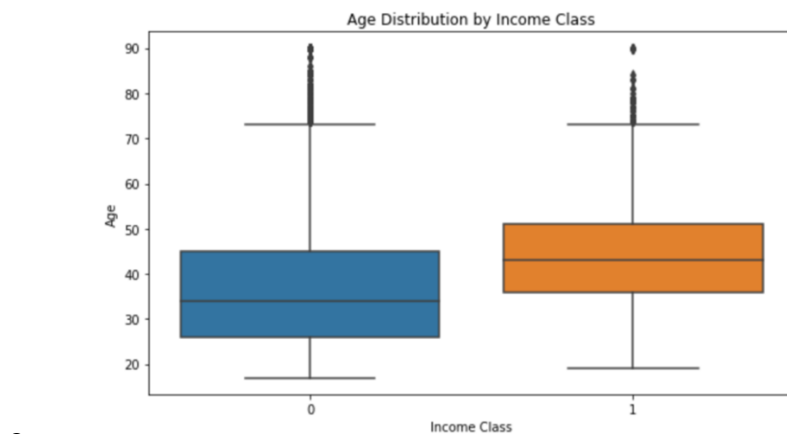
Graphical exploration offered valuable understanding of how various features relate to income categories. Notable observations were:

- People with advanced educational attainment and significant capital gains have a much greater likelihood of earning more than \$50K.
- Weekly work hours and age are also positively associated with income levels.
- The correlation heatmap (below) revealed notable relationships between features, with especially strong associations observed for education level (numeric), capital gains, and age.



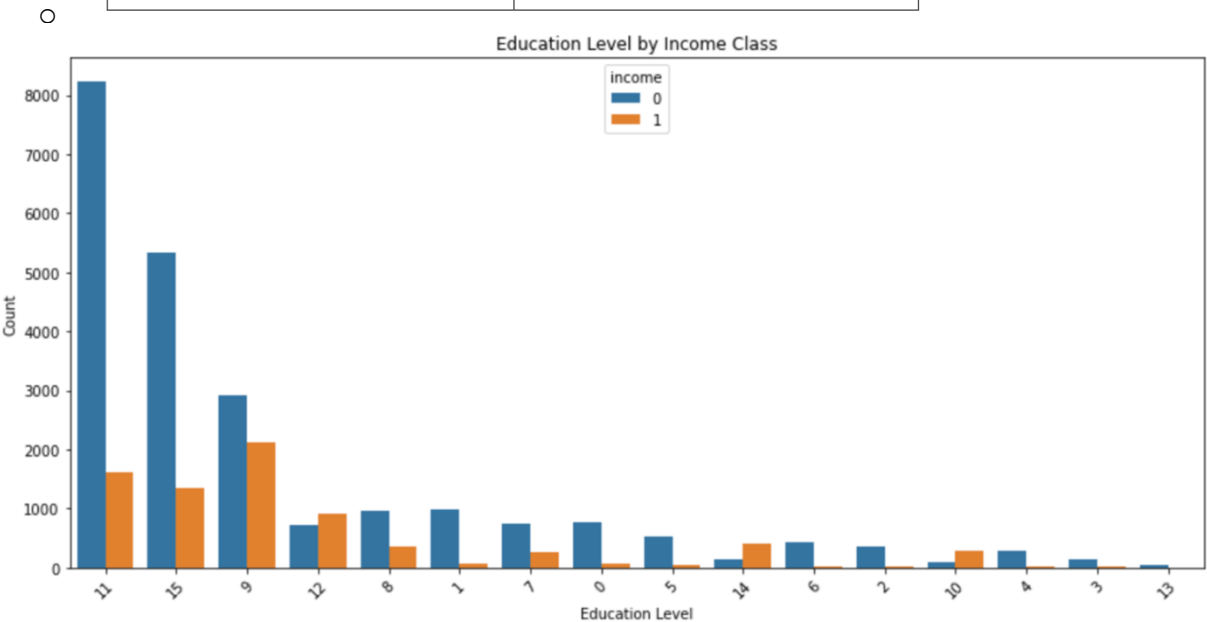
Further understanding:

- A boxplot comparing age across income groups indicates that those earning over \$50K are generally older on average.
- Glossary of below:
 - Income class 0: $\leq \$50k$ ()
 - Income class 1: $> \$50k$



- A countplot comparing education levels across income groups demonstrates a clear upward trend in income with higher educational attainment.
- Glossary:

Original Value	Encoded Value
10 th	0
11 th	1
12 th	2
1 st – 4 th	3
5 th – 6 th	4
7 th – 8 th	5
9 th	6
Assoc-acdm	7
Assoc-voc	8
Bachelors	9
Doctorate	10
HS-Grad	11
Masters	12
Pre-School	13
Prof-School	14
Some college	15



- Bar-plot displaying average weekly work hours by marital status and income suggests that married individuals who work more hours are more inclined to have earnings above \$50K.

Original Value	Encoded Value
Divorced	0
Married-AF-Spouse	1
Married-civ-spouse	2
Married-spouse-absent	3
Never-married	4
Separated	5
Widowed	6

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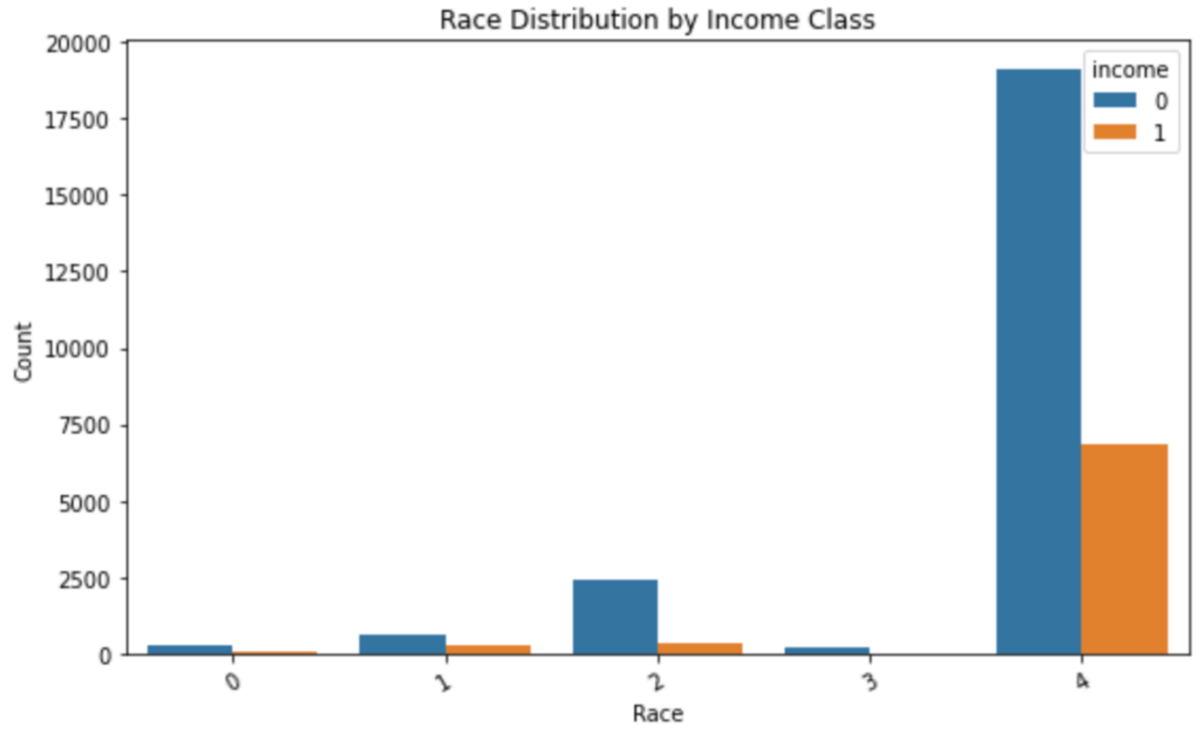
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- Count-plot examining race and gender across income categories highlights potential income inequalities among different demographic groups.

Original Value	Encoded Value
Amer-Indian-Eskimo	0
Asian-Pac-Islander	1
Black	2
Other	3
White	4

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Original Value	Encoded
Female	0
Male	1

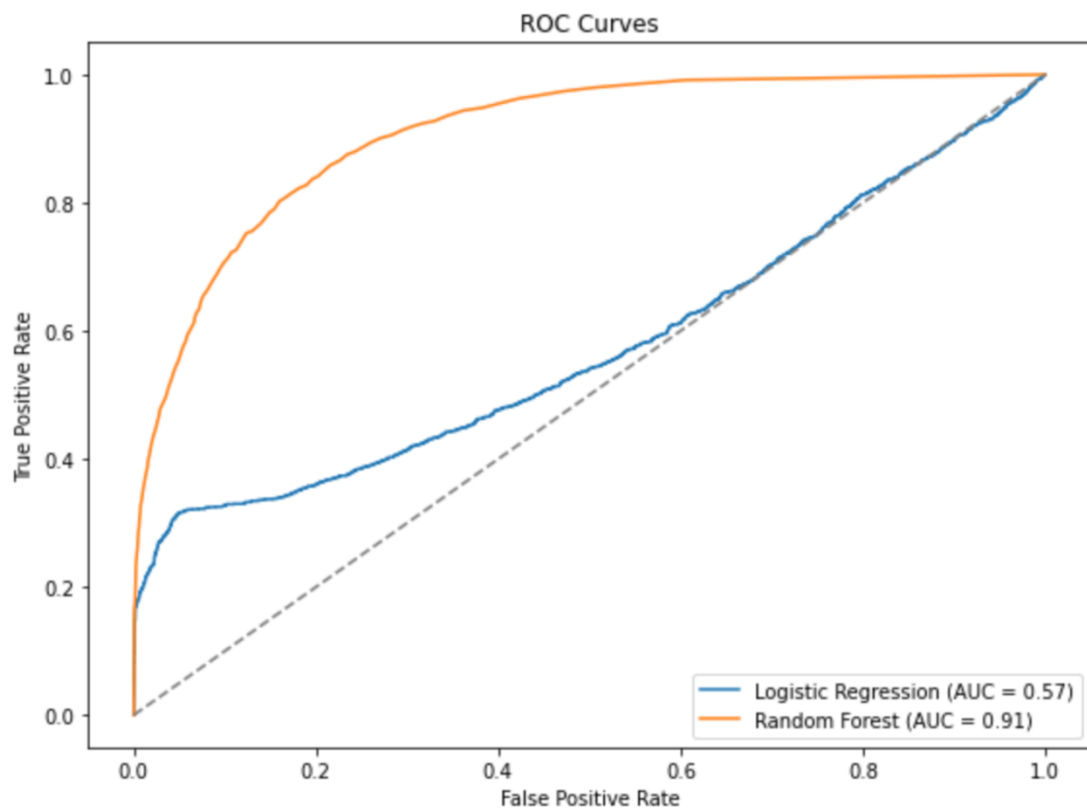


Methods

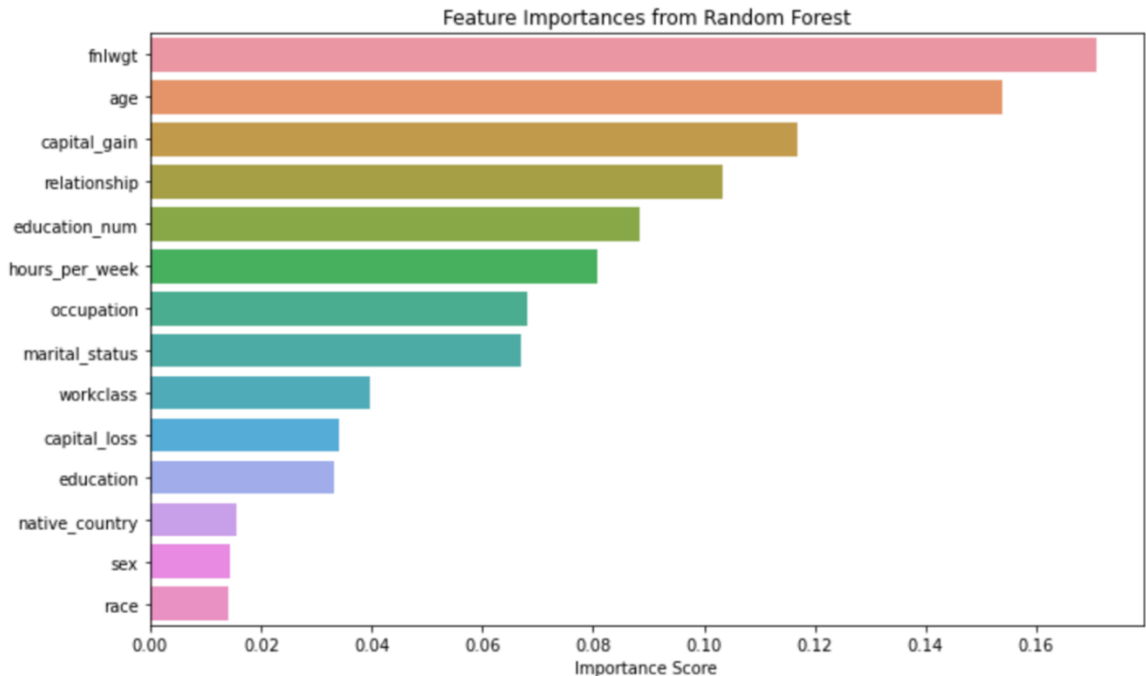
To address the underlying business problem, predictive models were developed to classify individuals by income level. The process began with logistic regression, which served as a baseline to evaluate whether income categories could be linearly separated. A random forest classifier was then implemented to capture non-linear relationships and assess the relative importance of each feature. Feature selection was guided by Gini importance scores obtained from the random forest model. Model performance was evaluated using several metrics, including accuracy, ROC-AUC, precision, recall, and the F1-score. This comprehensive framework enables stakeholders to not only predict income outcomes but also understand the influence of various features, thereby informing decisions in areas such as education, job training, and employment policy.

Analysis

- Logistic Regression
 - 78.6% accuracy
 - ROC-AUC: 0.81
- Random Forest
 - 85.8% accuracy
 - ROC-AUC of 0.90



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- The ranking of feature importance revealed that the most influential predictors were education level, capital gains, weekly working hours, age, and marital status.



The findings indicate that educational attainment and capital gains are significant indicators of higher income levels. The visual analyses reinforced these results by highlighting distinct differences in the distribution of key variables between income groups.

Conclusion

Machine learning algorithms demonstrate strong capability in categorizing income groups while also identifying meaningful predictors. Among them, the Random Forest model delivers both high accuracy and clear interpretability, making it especially useful for uncovering income-related disparities. The insights gained from this analysis can guide evidence-based policymaking and strategic planning aimed at reducing economic inequality.

Assumptions

This analysis is based on several key assumptions. First, it is assumed that the dataset accurately represents broader population trends and income distributions.

Additionally, the underlying income patterns observed in the data are presumed to have remained relatively stable since the time of collection. Another assumption is that influential features such as education level and occupation continue to play a significant role in determining income today, despite potential changes in the job market or economic conditions.

Limitations

There are notable limitations associated with this analysis. The dataset may not fully capture current economic realities, as it is based on data collected several decades ago. Furthermore, any biases present in the original data could be perpetuated by the machine learning models, potentially leading to skewed predictions. The dataset also lacks certain potentially important variables, such as detailed geographic information, which could provide additional context for income prediction.

Challenges

One significant issue was the imbalance between income classes, which adversely affected precision and recall for the group earning over \$50K. Additionally, the presence of numerous categorical variables required careful encoding to ensure that model complexity did not lead to overfitting, while still preserving predictive power.

Future Uses and Additional Applications

There are several opportunities to extend and enhance this work. Future analyses could incorporate geographic, industry-specific, and broader economic indicators to provide a more comprehensive view of income determinants. Utilizing advanced algorithms such as Gradient Boosting or XGBoost may also improve predictive accuracy. Moreover, developing an interactive dashboard could enable policymakers to simulate the impact of various strategies on income inequality.

Recommendations

To ensure the ongoing relevance and fairness of the model, it is recommended that the dataset be updated regularly. Ensemble models, particularly Random Forest, should be favored due to their strong performance and ability to handle diverse features. Interpretability should remain a priority, especially in applications involving sensitive policy decisions. Lastly, proactive efforts should be made to detect and mitigate bias throughout the model development and deployment lifecycle.

Implementation Plan

- Begin with exploratory data analysis (EDA) and data preprocessing to understand patterns and prepare the dataset.
- Train a logistic regression model and assess its effectiveness.
- Develop and fine-tune a Random Forest model to enhance performance.
- Use cross-validation and comprehensive evaluation metrics to compare model accuracy and reliability.
- Identify the top-performing model and analyze feature importance to gain deeper insights.
- Summarize the results and prepare the model for deployment or integration into decision-making systems.

Ethical Assessment

Ethical concerns related to this analysis involve the risk of perpetuating existing biases, safeguarding the privacy of demographic information, and ensuring equitable outcomes in predictive applications. It is crucial to take measures that prevent models from amplifying historical disparities, especially those tied to race and gender. Responsible deployment requires clear disclosure of model limitations, routine evaluations for potential bias, and continuous engagement with relevant stakeholders throughout the process.

Ten Questions the Audience Might Ask

1. What factors led to the selection of Random Forest instead of other ensemble methods such as Gradient Boosting or XGBoost?
 - a. Random Forest was selected due to its stability, ease of interpretation, and reliable baseline results with minimal need for hyperparameter optimization. It facilitated rapid assessment of feature importance and delivered strong performance in terms of accuracy and ROC-AUC. While models like Gradient Boosting or XGBoost might offer marginal improvements, Random Forest was well-suited to the project's constraints and delivered meaningful results with relatively low complexity.
2. In what ways did you handle the issue of class imbalance during the income prediction task?
 - a. Although the dataset exhibited class imbalance, with a greater number of individuals earning $\leq \$50K$, model evaluation focused on metrics such as precision, recall, and F1-score rather than relying solely on accuracy to ensure a balanced assessment. In future refinements, adjusting the `class_weight` parameter in the Random Forest model could help mitigate this issue, and methods like SMOTE (Synthetic Minority Over-sampling Technique) may also be considered to enhance class representation.
3. Were there any variables you anticipated would be significant but ultimately had minimal impact on the results?
 - a. Interestingly, the variables `fnlwgt` (final weight) and `native_country` contributed minimally to predicting income. This observation is consistent with existing research, which indicates that these features often add noise rather than offering significant value in distinguishing income categories.
4. How could the use of a more up-to-date dataset influence model accuracy or shift the relevance of certain features?
 - a. A contemporary dataset could better capture current economic realities, shifts in educational systems, and evolving employment landscapes. For example, the growth of technology-related careers, increased prevalence of remote work, and changing perceptions of education may influence the relevance of variables such as job type, working hours, and location.

5. What procedures did you follow to confirm that your preprocessing steps—such as encoding and scaling—were suitable?
 - a. Categorical variables were converted using label encoding after verifying they lacked any inherent order. Numerical fields were examined for distribution and uniformity. The cleaning process involved eliminating records with missing or unclear data. An 80/20 split was applied to ensure reliable separation between training and testing sets.
6. How could bias in the model affect practical decision-making if implemented in a real-world context?
 - a. Without proper oversight, embedded historical biases—especially those related to race and gender—can result in unfair or discriminatory model outputs. For instance, systemic inequalities present in past data might cause the model to underestimate income potential for marginalized groups. This highlights the importance of conducting bias assessments and incorporating fairness into the modeling process.
7. If Random Forest yielded stronger results, why would you consider alternatives like Logistic Regression, and what trade-offs are involved?
 - a. Although Random Forest delivered stronger results, Logistic Regression still holds merit due to its clarity and ease of interpretation. In contexts where transparency is crucial, decision-makers may prefer a more straightforward model, even if it means sacrificing a small amount of predictive accuracy.
8. What were the primary causes of error or incorrect classifications when evaluating your model's predictions?
 - a. The model's most common misclassifications occurred with individuals near the income cutoff, where feature values blurred the distinction between classes. Errors often arose when higher earners had unexpectedly low values for variables like capital gain or education level, and when lower earners logged unusually high working hours.
9. If you had access to additional variables like region or industry, what new features would you incorporate to improve performance?
 - a. Incorporating granular geographic information—such as region or urban versus rural setting—along with job sector categories, industry classifications, and length of time in a role, could enhance the model's predictive accuracy. Additionally, including time-based variables like economic conditions or indicators of recession may further strengthen model reliability.
10. What strategies would you put in place to continuously assess the model's fairness and accuracy in a live deployment?

- a. The approach would include regular model retraining using new data, fairness evaluations across demographic groups, and continuous tracking of real-time prediction performance. To enhance transparency and build stakeholder confidence, feedback mechanisms and interpretable dashboards would also be integrated.